

IMPORTANT QUESTIONS SET 1I

Subject: Engineering Chemistry BAS-202

NOTE: i) Attempt all sections. If require any missing data then choose suitably.

MM =70

SECTION A

1. Attempt all the following questions in brief

7x2= 14

Qno.	Question	CO
a.	Explain, which one will exhibit higher value of $\lambda_{\max}$ in UV/Visible spectra of CH_3COCH_3 and $\text{CH}_2=\text{CHCOCH}_3$	2
b.	Define the term Hypo chromic and Hyper chromic shift?	2
c.	Explain why we use dry ether in preparation of Grignard reagent.	5
d.	Classify the following as thermo plastic and thermosetting. Poly styrene, Polythene, Bakelite, Nylon, Urea formaldehyde, Teflon.	5
e.	Write down the absorption frequencies for following groups: i) $-\text{OH}$ ii) $>\text{C}=\text{O}$ iii) $-\text{CHO}$ iv) $-\text{CH}_3$	2
f.	What is gross and net calorific value?	4
g.	Comment on the electrical and lubrication property of graphite	1

SECTION B

2. Attempt any *three parts* of the following questions

3X7 = 21

Qno	Question	CO
a	i) What are the 12 principles and importance of green synthesis? ii) Write short notes on Plaster of Paris with applications.	1 3
b	i) What is battery? Discuss the working, construction and principle of Leclanche cell. ii) Discuss preparation, structures and properties of buckminsterfullerene.	3 1
c	i) Name and write structure of three modes of orientation of groups in polypropylene. Which of them is prepared by the Ziegler-Natta Catalyst? ii) Discuss the anodic coatings done for the protection of copper vessel.	4 3
d	i) Write down the molecular orbital configuration of NO , NO^+ and NO^- . Arrange them in increasing order of their stability ii) Discuss the green route of synthesis of paracetamol.	1
e	i) Discuss the structure, preparation, properties and applications of fullerene ii) 0.25 gm of a coal sample on burning in combustion chamber in presence of O_2 was found to increase weight of CaCl_2 tube by 0.08 gm and KOH tube by 0.5 gm. Find % of carbon and Hydrogen in the coal.	1 4
f	i) Name the raw material used in the preparation of Portland cement? Discuss the details of manufacturing and setting of Portland cement with schematic diagram. ii) Discuss the impressed current cathodic protection method	3
g	i) Explain the term chemical shift along with shielding and de-shielding. Give the number of ^1H NMR signals and their splitting pattern in the following compounds. (1) $(\text{CH}_3)_3\text{C}-\text{CO}-\text{CH}_3$ (2) $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{Cl}$ (3) $\text{C}_6\text{H}_5\text{CH}_3$ (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (5) CH_3CHCHO ii) Draw the Molecular Orbital diagram of HF molecule and comment on the stability and polarity of the covalent bond between H and F.	4 1

SECTION C

Qno 3	7X1=7	CO
	i) Explain the lime soda process for softening the hard water. Calculate the amount of lime and Soda required to soften 20,000 lit of water having following analysis $\text{Ca}(\text{HCO}_3)_2 = 3.50 \text{ ppm}$, $\text{Mg}(\text{HCO}_3)_2 = 6.28 \text{ ppm}$, $\text{CaSO}_4 = 3.92 \text{ ppm}$, $\text{MgSO}_4 = 7.4 \text{ ppm}$, $\text{MgCl}_2 = 2.9 \text{ ppm}$, ii) What are ion exchangers .With the help of neat sketch discuss ion exchange process for water softening . compare its merits over zeolite process.	4

Qno 4	7X1=7	CO
	i) Write the method of preparation and uses of the following polymers: Nylon 6, Perspex, Neoprene, Teflon and Darlon Poly styrene, Bakelite. ii) Write down the molecular orbital configuration of NO, NO^+ and NO^- .Arrange them in increasing order of their stability	5 1

Qno5	7X1=7	CO
	i) Discuss the different types of boiler troubles causes by hard water. ii) Calculate the amount of lime & soda required for softening 30000.liters of water using 20 ppm of sodium aluminate as coagulant. Impurities in water are as follows $\text{Ca}^{2+} = 160 \text{ ppm}$,$\text{Mg}^{2+} = 96 \text{ ppm}$, dissolved $\text{CO}_2 = 34 \text{ ppm}$ and $\text{HCO}_3^- = 403 \text{ ppm}$.	4
Qno6	7X1=7	CO
	i) Write the chemical structure of poly isoprene. How would you crosslink the chains of polyisoprene. ii) What are conducting polymers? Classify conducting polymers and mention their important applications.	5
Qno 7	7X1=7	CO
	i) For XY_2 bent molecule shows various types of stretching and bending vibrations in I R spectroscopy. ii) What is Beer-Lambert law in UV-VIS absorption spectroscopy? A compound having concentration 10^{-3} g/l resulted absorbance value 0.20 at $\lambda_{\text{max}} 510 \text{ nm}$ using 1 cm cell . Calculate its absorptivity and molar absorptivity values. Molecular weight of compound is 400.	2

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